

Python

① What is Python?

- Python was created by Guido van Rossum, and released in 1991.
- It is general-purpose object oriented programming language.
- It is platform independent. widely used in ML, AI.
- It is easy to learn and require less code to develop the applⁿ.

② Why Python?

- Python is Interpreted, object-oriented, high level language & it is compatible with different platforms like windows, Mac, Linux.
- Python has a simple syntax as compared to other languages.
- It requires less line of code and consist of many libraries & inbuilt function that make it easy to maintain & portable.

③ Applications of Python.

- web and Internet Development.
- Game.
- Operating systems.
- GUI based desktop applⁿ.

④ Advantage of Python

- Easy to use.
- Interpreted language.
- It is free and open source.
- Extensible.
- Object-oriented.
- It has built-in data structure.

⑤ What is PEP 8?

Ans PEP 8 is a coding convention, a set of recommendations about how to write your python code more readable.

⑥ What is pickling and unpickling?

Ans - Pickling is the process whereby a Python object hierarchy is converted into a byte stream, and "unpickling" is the inverse operation, whereby a byte stream (from a binary file or bytes-like object) is converted back into an object hierarchy.

⑦ How is Python interpreted?

Ans - Python is an interpreted language. It does not require prior compilation of code and executes instructions directly.

⑧ What do you mean by Python literals?

Ans Literals can be defined as a data which is given in a variable or constant. Python supports following literals.

→ String literals, Numeric, Boolean, Special like

⑨ How is memory managed in Python?

Ans Python uses the dynamic memory allocation which is managed by the Heap data structure.

(10) what are the tools that help to find bugs or perform the static analysis

Ans - Pychecker is a tool that detects the bugs in Python source code and Pylint is a tool that verifies the module meets the coding standard.

(11) What are Python decorators?

Ans Python decorator is a specific change that we make in Python syntax to alter funⁿ easily.

(12) diff b/w list & tuple?

<u>Ans</u> - list	<u>tuple</u>
• list is mutable • consumes more memory • many built-in methods are available.	• tuple is immutable. • consumes less memory. • Does not have many built-in methods.

Node
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(13) How are arguments passed by value or by reference?

Ans - Everything in Python is an object, and all variables hold references to the objects. Therefore, you cannot change the value of the references. However, you can change the objects if it is mutable.

(14) What are Dict and List comprehensions - are?

Ans - They are syntax constructions to make easy creation of dictionary or list.

(15) What are built-in types does Python provide?

Ans
Mutable
List, Dictionaries, ~~Strings~~
Set, ~~tuples~~, ~~Numbers~~

Immutable
Strings, tuples, Numbers.

Ans

(17) What is lambda in Python? → It is an anonymous funⁿ which
except many no. of argument but
Ans - It is a single expression anonymous function return in a
often used as inline function. single line
express

(18) Why lambda forms in Python do not have statement?
Ans - ~~It~~ ^{lambda} does not have statement because it is used
to make new function object and then return
them at runtime.

(19) Explain pass in Python?

Ans - Pass means no-operation Python-statement.
we generally use it as a placeholder.

(20) In Python what are iterators?

Ans - Iterators are used to iterate a group of element,
container like a list.

(21) What is slicing in Python?

Ans - A mechanism to select a range of items from
sequence types like list, tuple, string is
known as slicing.

(22) What are generators in Python?

Ans - The way of implementing iterators are
known as generators.

(23) What is docstring in Python?

Ans A Python documentation string is known as docstring. It is a way of documenting Python funⁿ, modules & cla

(24) What is the diff b/w xrange and range?

Ans xrange returns the xrange object while range returns the list and uses the same memory and no matter what the range size is.

(25) What is module and package in Python?

Ans Module is the way to structure a program. The folder of Python program is a package of module.

(26) Use of the split function in Python?

Ans - The use of the split funⁿ in Python is that it breaks a string into shorter strings using the defined separator. It gives a list of all words present in the string.

(27) What is enumerate() in Python?

Ans - enumerate() in Python is a built-in function used for assigning an index to each item of the iterable object.

(28) Membership operators.

Ans These operators test for membership in a sequence such as lists, strings, or tuples. Two membership operators are used in Python. (in, not in).

(29) How can you reverse array in Python?

Ans - You can use reverse().

30) How to overload constructors or methods in Python?

Ans - Python's constructor: `__init__()` is the first method of a class. Whenever we try to instantiate an object `__init__()` is automatically invoked by python to initialize members of an object.

31) differⁿ b/w `remove()` funⁿ & `del` statement?

Ans - The user can use the `remove()` funⁿ to delete a specific object in the list.

If you want to delete an object at a specific location in the list, you can either use `del` or `pop`.

32) Why do we use `join()` funⁿ in python?

Ans - The `join()` is defined as a string method which returns a string value. It provides a flexible way to concatenate the strings.

33) Operator in Python?

Ans - Arithmetic, Assignment, Logical, Membership, Identity, Bitwise.

34) What is dictionary in Python?

Ans - The Python dict is a built-in data type. It defines a one-to-one relationship b/w keys and values.

35) differⁿ b/w

Criteria	Java	Python
Easy of use	Good	Very Good
coding speed	Average	Excellent
Data type	Static type	Dynamic type
Data science and ML appl ⁿ	Average	Very Good

Q36) What is type conversion in Python?

Ans Type conversion refers to the conversion of one data type into another.

- int()
- hex()
- set()
- str()
- float()
- oct()
- list()
- ord()
- tuple()
- dict()

Q37) What is diff b/w Array and list?

Ans Array can hold only a single data type elements whereas lists can hold any data type elements.

Q38) What is `__init__`?

Ans The `__init__` is a method or constructor in Python. This method is automatically called to allocate memory when a new object / instance of a class is created. All classes have the `__init__` method.

Q39) What is `self` in Python?

Ans - `self` is an instance or an object of a class. The `self`-variable in the `__init__` method refers to the newly created object while in other methods, it refers to the object whose method was called.

Q40) What is PYTHONPATH?

Ans - PYTHONPATH is an environment variable which is used when a module is imported. The interpreter uses it to determine which module to load.